

CodeAlpha Project Management Tool - Master Guide

1. What You Built

You built a full-stack Project Management Tool using React, Node.js, Express and MongoDB. Users can create projects, view projects, update project status and delete projects. This is not just a frontend project; it is a complete CRUD application.

2. Architecture Diagram

```
graph TD
    User --> ReactUI[React UI]
    ReactUI --> AxiosService[Axios Service]
    AxiosService --> ExpressRoute[Express Route]
    ExpressRoute --> Controller
    Controller --> MongooseModel[Mongoose Model]
    MongooseModel --> MongoDB
```

User
↓
React UI
↓
Axios Service
↓
Express Route
↓
Controller
↓
Mongoose Model
↓
MongoDB

Response returns back through the same chain.

3. Backend Folder Structure

```
backend/  
config/db.js  
models/Project.js  
controllers/projectController.js  
routes/projectRoutes.js  
server.js
```

server.js starts Express and connects MongoDB.
Routes receive requests.
Controllers contain logic.
Models talk to MongoDB.

4. Frontend Folder Structure

```
frontend/src/  
components/ProjectForm.jsx  
components/ProjectList.jsx  
pages/Dashboard.jsx  
services/projectService.jsx
```

Dashboard = page controller.
ProjectForm = user input.

ProjectList = display projects.
projectService = API communication.

5. Every API Endpoint

POST /api/projects
Create a project.

GET /api/projects
Fetch all projects.

DELETE /api/projects/:id
Delete a project.

PUT /api/projects/:id
Update project status.

6. Important Functions

Project.create()
Creates document.

Project.find()
Returns all projects.

Project.findByIdAndDelete()
Deletes project.

Project.findByIdAndUpdate()
Updates project.

7. React Concepts Learned

useState()
Stores data inside component.

useEffect()
Runs code after component loads.

Props
Used to pass data/functions between components.

Component Architecture
Breaking UI into reusable pieces.

8. Axios Concepts Learned

axios.get()
Fetch data.

axios.post()
Create data.

axios.put()
Update data.

axios.delete()
Delete data.

9. Errors You Encountered

1. Cannot find module server.js

Reason: Wrong folder.

2. MongoDB connection issues.

Reason: Database/server setup.

3. Cannot GET /api/projects

Reason: Route not connected correctly.

4. Identifier already declared.

Reason: Duplicate code in route file.

5. JSON parse errors.

Reason: Invalid request format.

6. Cast to ObjectId failed.

Reason: You literally used PASTE_ID_HERE instead of real MongoDB id.

10. Interview Story

I developed a full-stack Project Management Tool using React, Express and MongoDB.

I designed REST APIs for Create, Read, Update and Delete operations.

I integrated Axios on the frontend and connected it with Express routes.

MongoDB was used for persistent data storage through Mongoose models.

I also debugged routing, API and database issues during development.

11. Questions Interviewers May Ask

Explain CRUD.

Explain REST API.

Why React?

Why MongoDB?

Difference between SQL and NoSQL?

What is Axios?

What is useEffect?

What is useState?

What is Express?

What is Mongoose?

Explain your project architecture.

12. Memory Map

Remember this one line:

Frontend collects data.
Service sends request.
Route receives request.
Controller executes logic.
Model accesses database.
MongoDB stores data.

If you understand this chain, you understand the entire project.